

Topic 8

Probability

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

Learning focus

Explain every method before practising. Use the original practice questions to check calculation, reasoning and communication.

Topic map

8.1	Probability scale and complements
8.2	Relative frequency and sample spaces
8.3	Combined independent events
8.4	Tree diagrams and dependent events
8.5	Venn diagrams and conditional reasoning

Core foundations and Extended applications are integrated throughout. Study each example, then attempt the practice paper without the notes.

1. Single events

A probability lies between 0 and 1. For equally likely outcomes, $P(A) = \text{favourable outcomes} / \text{total outcomes}$. Complement $P(\text{not } A) = 1 - P(A)$. Experimental relative frequency approximates probability when trials are numerous, but is not guaranteed to match exactly.

Worked example

A fair die gives $P(\text{even}) = 3/6 = 1/2$; therefore $P(\text{odd}) = 1/2$.

Exam check: Check that the probabilities of mutually exclusive exhaustive outcomes add to 1.

2. Sample space and combinations

List the full sample space systematically using a table or tree. For either event A or event B, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. If events cannot occur together, the overlap is zero. With equally likely pair outcomes, count ordered pairs rather than just distinct totals.

Worked example

Two fair dice have 36 ordered outcomes. Sum 7 occurs in six pairs, so $P(\text{sum } 7) = 6/36 = 1/6$.

Exam check: (2,5) and (5,2) count separately when dice are distinguishable.

3. Independent events

For independent events, knowing one does not change the probability of the other. Multiply along tree branches and add probabilities of alternative paths. Replacement leaves the original probabilities unchanged; without replacement they change.

Worked example

A fair coin is tossed twice: $P(\text{two heads}) = 1/2 \times 1/2 = 1/4$; $P(\text{exactly one head}) = 1/4 + 1/4 = 1/2$.

Exam check: Check whether the item is replaced before drawing the second branch.

4. Dependent events and conditional information

Without replacement, both numerator and denominator may change after the first selection. $P(A \text{ given } B)$ uses the reduced group satisfying B as its denominator. A Venn diagram can locate that group: $P(A|B) = n(A \cap B) / n(B)$ for equally likely people.

Worked example

Of 12 pupils, 5 play chess, 4 music and 2 both. Among music pupils, $P(\text{chess} | \text{music}) = 2/4 = 1/2$.

Exam check: Do not divide by the full class when the wording says "given that" or "among those".

5. Reading trees and checks

The two outcomes at a branch sum to 1 at each split. Complete a tree before multiplying paths. If an event has several separate paths, add their path probabilities. As a final check, all terminal path probabilities must sum to 1.

Worked example

A bag contains 3 red and 2 blue counters; without replacement $P(\text{two red}) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$.

Exam check: A second draw probability may use four remaining counters, not five.

Before the practice paper

Method and accuracy

Write the formula or first operation, show the intermediate step, give units when needed and check by substitution, estimation or a sketch. On non-calculator work, keep exact forms when requested.

Practice paper

Topic 8 · Probability

Instructions

Original Cambridge-style practice. Answer all questions. Section A is multiple choice; Section B requires working and final answers. Try Section A without a calculator; use a calculator in Section B only where needed.

Candidate name:	
Class / date:	

Total: 32 marks

Section A · Multiple choice

Choose one answer for each question.

1. $P(A) = 0.35$. What is $P(\text{not } A)$? [1]

- A 0.35
- B 0.65
- C 1.35
- D 0.75

2. $P(\text{heads twice})$ for a fair coin? [1]

- A $1/2$
- B $1/4$
- C $1/3$
- D $1/8$

3. Probability of a 6 on a fair die? [1]

- A $1/3$
- B $1/6$
- C $1/4$
- D 6

4. A certain event has probability [1]

- A 0
- B 0.5
- C 1
- D 6

5. For independent events, $P(A \text{ and } B)$ equals [1]

- A $P(A)+P(B)$
- B $P(A)P(B)$
- C $P(A)/P(B)$
- D $1-P(A)$

6. A bag has 2 red and 3 blue. $P(\text{red})$ is [1]

- A $2/3$
- B $3/5$
- C $2/5$
- D $1/5$

7. With 3 red and 2 blue, $P(\text{red then red})$ without replacement? [1]

- A $9/25$
- B $3/10$
- C $1/5$
- D $2/5$

8. If $P(A)=0.5$, $P(B)=0.4$ and $P(A \cap B)=0.1$, find $P(A \cup B)$. [1]

- A 0.1
- B 0.8
- C 0.9
- D 1.0

Section B · Structured questions

9. A fair coin is tossed and a fair six-sided die is rolled.

(a) Find $P(\text{heads and six})$. [2]

(b) Find $P(\text{tails and an even number})$. [2]

(c) Find $P(\text{heads or tails})$. [1]

[Total: 5]

10. A bag contains 4 red and 3 blue counters. Two are selected without replacement.

(a) Find $P(\text{two red})$. [2]

(b) Find $P(\text{red then blue})$. [2]

(c) Find $P(\text{one of each colour})$.

[3]

[Total: 7]

11. Of 30 pupils, 16 take French, 13 take Spanish and 5 take both.

(a) Find how many take neither.

[2]

(b) Find $P(\text{French or Spanish})$.

[2]

(c) Find $P(\text{French given Spanish})$.

[2]

[Total: 6]

12. A biased spinner lands on A with probability 0.30, B with 0.45 and C otherwise. It is spun twice independently.

(a) Find $P(C)$.

[1]

(b) Find $P(\text{two C outcomes})$.

[2]

(c) Find $P(\text{exactly one A})$.

[3]

[Total: 6]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Award method marks when the mathematical setup is correct even if an arithmetic slip follows.
Check each part independently.

Section A · Answers and reasons

1. B — 0.65. Use the complement: $1 - 0.35 = 0.65$.
2. B — $1/4$. Two independent heads probabilities multiply: $(1/2)^2$.
3. B — $1/6$. One favourable face among six equally likely faces.
4. C — 1. A certain event occurs on every outcome.
5. B — $P(A)P(B)$. Independent events multiply for the intersection.
6. C — $2/5$. Two of five counters are red.
7. B — $3/10$. Without replacement: $3/5 \times 2/4 = 3/10$.
8. B — 0.8. Use inclusion-exclusion: $0.5 + 0.4 - 0.1 = 0.8$.

Section B · Worked solutions

9. a) $(1/2)(1/6) = 1/12$ [2]. b) $(1/2)(3/6) = 1/4$ [2]. c) A coin always shows heads or tails, so 1 [1].

10. a) $(4/7)(3/6) = 2/7$ [2]. b) $(4/7)(3/6) = 2/7$ [2]. c) red-blue or blue-red = $(4/7)(3/6) + (3/7)(4/6) = 4/7$ [3].

11. a) Union = $16 + 13 - 5 = 24$, neither = 6 [2]. b) $24/30 = 4/5$ [2]. c) $5/13 = 5/13$ [2].

12. a) $1 - 0.30 - 0.45 = 0.25$ [1]. b) $0.25^2 = 0.0625$ [2]. c) A then not A or not A then A: $2(0.3)(0.7) = 0.42$ [3].

Review your errors

Mark the first wrong step: setup, algebra, arithmetic, units, graph or reasoning. Redo that part without seeing the solution.

Syllabus reference: Cambridge IGCSE Mathematics 0580, 2025-2027,
www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf. Paper style informed by the June 2026 papers and
examiner report provided by the user. All practice questions are original.